

A Spherical Finite-Difference Model of Fission-Product Diffusion in TRISO Fuel Kernels for Fluoride-Salt-Cooled High-Temperature Reactors

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Abstract

Tristructural-isotropic (TRISO) particles are the reference fuel form for several advanced reactor concepts, including the fluoride-salt-cooled high-temperature reactor (FHR). Predicting the release of gaseous and metallic fission products from the fuel kernel is central to source-term assessment and to demonstrating the integrity of the coating layers. This work formulates the migration of a fission-product species through a spherical TRISO kernel as a transient diffusion problem governed by Fick's second law. Exploiting the spherical symmetry of the kernel reduces the governing equation to one radial dimension. The reduced equation is discretised with a forward-time, centred-space (FTCS) finite-difference scheme on a uniform radial mesh, yielding an explicit update for the nodal concentrations. Symmetry at the kernel centre and a zero-flux condition at the outer node close the system, with the centre singularity removed through a L'Hopital limit that converts the governing operator to $3D \partial^2 c / \partial r^2$. The framework admits optional source terms representing in-pile fission-product generation, radioactive decay, and grain-boundary trapping, and supports both a constant diffusion coefficient and a temperature-dependent coefficient. The resulting model provides a transparent, low-cost baseline against which higher-fidelity solvers and benchmark solutions can be compared. Numerical results and verification against analytical limits are reported separately.

Keywords: TRISO fuel; fission-product transport; Fick's second law; finite-difference method; FTCS scheme; fluoride-salt-cooled high-temperature reactor.

Nomenclature

Symbol	Definition	Units
c	Concentration of the diffusing fission-product species	mol m^{-3}
C_i^j	Discrete concentration at node i , time level j	mol m^{-3}
D	Diffusion coefficient	$\text{m}^2 \text{s}^{-1}$
D_0	Arrhenius pre-exponential factor	$\text{m}^2 \text{s}^{-1}$
Q	Activation energy for diffusion	J mol^{-1}
R_g	Universal gas constant	$\text{J mol}^{-1} \text{K}^{-1}$
T	Absolute temperature	K
r	Radial coordinate	m
R	Kernel (outer) radius	m
t	Time	s
Δr	Radial mesh spacing	m
Δt	Time-step size	s
i	Radial node index (0 = centre, 4 = surface)	—
j	Time-level index	—
Fo	Mesh Fourier number, $D\Delta t/\Delta r^2$	—
Σ_f	Macroscopic fission cross section	m^{-1}
ϕ	Neutron flux	$\text{m}^{-2} \text{s}^{-1}$
Y	Fission-product yield	—
λ	Radioactive decay constant, $\ln 2/t_{1/2}$	s^{-1}
$t_{1/2}$	Half-life	s
κ	Grain-boundary trapping rate	s^{-1}
S	Volumetric source/sink term	$\text{mol m}^{-3} \text{s}^{-1}$

1 Introduction

A TRISO particle confines a fissile kernel within successive coating layers that act as a miniature pressure vessel for the fission products generated during irradiation. In a fluoride-salt-cooled high-temperature reactor (FHR), large numbers of these particles are dispersed in a graphitic matrix and cooled by molten salt, so the ability of each kernel and its coatings to retain fission products sets the radiological source term of the whole core [? ?]. Quantifying how a fission-product species migrates outward from the kernel is therefore a prerequisite for safety analysis and for interpreting post-irradiation examination data. At the level of a single kernel, the transport of a diffusing species is described by Fick’s second law [?]. The reduction of kernel-scale release to an equivalent-sphere diffusion problem has a long history in fuel modelling, dating to the classical treatments of fission-gas release [? ?] and continuing in modern TRISO performance codes [?]. The kernel is well approximated as a sphere, and for a species that is generated and redistributed uniformly the concentration field depends only on the radial coordinate. This reduction makes the problem tractable with elementary numerical methods while retaining the essential physics of curvature-weighted radial diffusion. The present paper sets out a compact model for this process. The continuous governing equation is reduced to one radial dimension in section 2 and discretised with an explicit finite-difference scheme in section 3. The boundary treatment, including the removal of the coordinate singularity at the kernel centre, is given in section 4. Section 5 catalogues the physical source and sink terms that can be appended to the base model, namely fission-product generation, radioactive decay, and grain-boundary trapping, together with the choice between a constant and a temperature-dependent diffusion coefficient. Section 6 defines the verification strategy and the planned numerical experiments, and sections 7 and 8 discuss the limitations of the explicit scheme and outline extensions toward an implicit,

Crank-Nicolson formulation. The aim is a baseline solver of deliberately modest complexity. Such a baseline serves two purposes: it exposes the discretisation choices in a form that can be audited line by line, and it provides reference output against which more elaborate finite-element or multilayer models can be benchmarked.

2 Model Formulation

2.1 Governing equation

The evolution of the molar concentration $c(\mathbf{r}, t)$ of a diffusing fission-product species is governed by Fick's second law,

$$\frac{\partial c}{\partial t} = D \nabla^2 c, \quad (1)$$

where D is the diffusion coefficient of the species in the kernel material. In spherical coordinates (r, θ, φ) the Laplacian expands to

$$\frac{\partial c}{\partial t} = D \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial c}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 c}{\partial \varphi^2} \right]. \quad (2)$$

2.2 Reduction to one dimension

Two modelling assumptions reduce eq. (2) to a radial problem.

Assumption 1 (uniformity). The kernel composition and the species source are uniform over each spherical shell, so the concentration carries no dependence on the polar or azimuthal angles. The θ and φ derivatives in eq. (2) vanish.

Assumption 2 (surface-coupled boundary). The concentration at any radial position is referenced to the value imposed at the kernel surface, which fixes the outer boundary condition and removes any need to resolve angular transport at the edge of the domain.

With angular dependence eliminated, the governing equation becomes

$$\frac{\partial c}{\partial t} = D \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) \right]. \quad (3)$$

2.3 Expanded radial form

Applying the product rule to the conservative flux term in eq. (3) gives

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c}{\partial r} \right) = \frac{1}{r^2} \left[2r \frac{\partial c}{\partial r} + r^2 \frac{\partial^2 c}{\partial r^2} \right] = \frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r}. \quad (4)$$

The governing equation therefore takes the simplified radial form used throughout the remainder of this paper,

$$\boxed{\frac{\partial c}{\partial t} = D \left(\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \right)}. \quad (5)$$

The first term on the right is ordinary one-dimensional diffusion. The second term is a curvature contribution proportional to $1/r$; it accounts for the shrinking spherical area available for transport

as the species moves toward the centre and becomes singular at $r = 0$, a feature handled explicitly in section 4.

3 Discretisation

3.1 Mesh and notation

The kernel radius is divided into a uniform mesh of spacing Δr . Radial nodes are indexed by i and time levels by j , so that the discrete concentration is $C_i^j \approx c(r_i, t_j)$ with $r_i = i \Delta r$. The illustrative configuration used here takes five nodes, $i = 0, \dots, 4$, corresponding to four concentric shells, with $i = 0$ at the centre ($r_0 = 0$) and $i = 4$ at the outer surface. Numbering from zero places the centre node exactly at the origin and keeps the curvature weight in the clean form $1/i$ introduced below.

3.2 Forward-time, centred-space scheme

The right-hand side of eq. (5) is discretised with central differences. The second derivative uses the standard three-point stencil,

$$\frac{\partial^2 c}{\partial r^2} \approx \frac{C_{i-1}^j - 2C_i^j + C_{i+1}^j}{\Delta r^2}, \quad (6)$$

and the first derivative uses the two-point centred stencil,

$$\frac{\partial c}{\partial r} \approx \frac{C_{i+1}^j - C_{i-1}^j}{2 \Delta r}. \quad (7)$$

The time derivative on the left-hand side is advanced with the forward (explicit) Euler difference,

$$\frac{\partial c}{\partial t} \approx \frac{C_i^{j+1} - C_i^j}{\Delta t}. \quad (8)$$

Substituting eqs. (6) to (8) into eq. (5) gives the fully discrete balance,

$$\frac{C_i^{j+1} - C_i^j}{\Delta t} = D \left(\frac{C_{i+1}^j - 2C_i^j + C_{i-1}^j}{\Delta r^2} + \frac{2}{r_i} \frac{C_{i+1}^j - C_{i-1}^j}{2 \Delta r} \right). \quad (9)$$

3.3 Explicit update

Solving eq. (9) for the unknown future value C_i^{j+1} yields the explicit time-marching rule,

$$C_i^{j+1} = C_i^j + \frac{D \Delta t}{\Delta r^2} (C_{i+1}^j - 2C_i^j + C_{i-1}^j) + \frac{D \Delta t}{r_i \Delta r} (C_{i+1}^j - C_{i-1}^j). \quad (10)$$

Collecting the three nodal values into stencil coefficients gives the form that is most convenient to implement,

$$C_i^{j+1} = C_i^j \left(1 - 2 \frac{D \Delta t}{\Delta r^2} \right) + C_{i+1}^j \left(\frac{D \Delta t}{\Delta r^2} + \frac{D \Delta t}{\Delta r^2} \frac{1}{i} \right) + C_{i-1}^j \left(\frac{D \Delta t}{\Delta r^2} - \frac{D \Delta t}{\Delta r^2} \frac{1}{i} \right), \quad (11)$$

where the substitution $r_i = i \Delta r$ has been used so that the curvature term reduces to a factor $1/i$. The interior nodes therefore differ only through this $1/i$ weight: $1/i = 1, \frac{1}{2}, \frac{1}{3}$ at $i = 1, 2, 3$ respectively. This is the quantity evaluated inside the spatial loop of any implementation.

3.4 Stability

The explicit scheme of eq. (10) is conditionally stable. The mesh Fourier number,

$$\text{Fo} = \frac{D\Delta t}{\Delta r^2}, \quad (12)$$

controls the leading diffusion stencil, whose central coefficient is $1 - 2\text{Fo}$. For a planar diffusion stencil the von Neumann condition requires $\text{Fo} \leq \frac{1}{2}$; the additional curvature term tightens this bound near the centre, so the time step must be chosen conservatively. An unstable choice of Δt produces growing oscillations rather than the smooth relaxation expected of a diffusion process, and this failure mode is one of the diagnostics checked in section 6.

4 Initial and Boundary Conditions

4.1 Initial condition

The kernel is taken to start from a quiescent state in which the concentration field is stationary,

$$\left. \frac{\partial c}{\partial t} \right|_{t=0} = 0, \quad (13)$$

so the prescribed initial profile (uniform unless stated otherwise) is held fixed until the source terms or boundary values drive the transient.

4.2 Centre node: symmetry and the removable singularity

At the kernel centre the spherical symmetry of the problem forbids any net radial flux,

$$\left. \frac{\partial c}{\partial r} \right|_{r=0} = 0. \quad (14)$$

Introducing a ghost node C_{-1} just inside the centre and applying the centred first difference of eq. (7) gives

$$0 = \frac{C_1 - C_{-1}}{2 \Delta r} \implies C_{-1} = C_1. \quad (15)$$

Substituting $C_{-1} = C_1$ into the second-derivative stencil collapses the two neighbour contributions,

$$\left. \frac{\partial^2 c}{\partial r^2} \right|_{i=0} = \frac{C_1 - 2C_0 + C_{-1}}{\Delta r^2} = \frac{2C_1 - 2C_0}{\Delta r^2}. \quad (16)$$

The curvature term $\frac{2}{r} \partial c / \partial r$ in eq. (5) is indeterminate at $r = 0$ because both numerator and the $1/r$ factor diverge in opposite senses. Its value follows from L'Hopital's rule,

$$\lim_{r \rightarrow 0} \frac{2}{r} \frac{\partial c}{\partial r} = \lim_{r \rightarrow 0} 2 \frac{\partial c / \partial r}{r} = 2 \left. \frac{\partial^2 c}{\partial r^2} \right|_{r=0}, \quad (17)$$

so that the full right-hand-side operator at the centre simplifies to

$$\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \xrightarrow{r \rightarrow 0} \frac{\partial^2 c}{\partial r^2} + 2 \frac{\partial^2 c}{\partial r^2} = 3 \frac{\partial^2 c}{\partial r^2}. \quad (18)$$

The governing equation at the centre node is then

$$\left. \frac{\partial c}{\partial t} \right|_{i=0} = 3D \left. \frac{\partial^2 c}{\partial r^2} \right|_{i=0}, \quad (19)$$

and combining eqs. (16) and (19) with the forward Euler step gives the explicit update for the centre node,

$$C_0^{j+1} = C_0^j + 6 D \Delta t \frac{C_1^j - C_0^j}{\Delta r^2}. \quad (20)$$

The factor of six arises from the product of the curvature factor of three in eq. (18) and the factor of two in the collapsed stencil of eq. (16).

4.3 Outer node: zero-flux (insulated) surface

At the outer node $i = 4$ the nodes adopt a zero-gradient (insulated) condition,

$$\left. \frac{\partial c}{\partial r} \right|_{r=R} = 0, \quad (21)$$

which, with an exterior ghost node C_5 , gives $C_5 = C_3$ by the same construction as eq. (15). The second-derivative stencil at the surface then collapses to

$$\left. \frac{\partial^2 c}{\partial r^2} \right|_{i=4} = \frac{C_5 - 2C_4 + C_3}{\Delta r^2} = \frac{2C_3 - 2C_4}{\Delta r^2}. \quad (22)$$

The curvature term needs separate care, and its treatment differs fundamentally from the centre. The surface lies at a finite radius $r = R$, so the factor $2/r = 2/R$ is an ordinary finite number and the product $\frac{2}{r} \partial c / \partial r$ is not an indeterminate form. The L'Hopital argument of eq. (17), which produced the factor-of-three enhancement at the centre, therefore does not apply at the surface. With the insulated condition $\partial c / \partial r|_R = 0$ the curvature term is simply $\frac{2}{R} \times 0 = 0$, so only the second-derivative stencil survives. The explicit update at the surface consequently carries a factor of two rather than the factor of six found at the centre,

$$C_4^{j+1} = C_4^j + 2 D \Delta t \frac{C_3^j - C_4^j}{\Delta r^2}. \quad (23)$$

The insulated surface is the natural choice for verifying conservation, since it isolates the kernel and the total quantity of species must then remain constant in the absence of source terms. A Dirichlet condition that fixes C_4 to a prescribed surface concentration (Assumption 2) is substituted in its place when coupling to the coating layers is required.

4.4 Interior-node summary

Collecting the results, the interior nodes $i = 1, 2, 3$ advance by eq. (11) with the curvature weight $1/i$, while the centre node $i = 0$ uses eq. (20) with its factor-of-six enhancement and the insulated surface node $i = 4$ uses eq. (23) with a factor of two. The two end-node updates are therefore not identical: the factor of six is specific to the L'Hopital limit at the singular centre, whereas the finite-radius surface retains the ordinary factor of two. The complete nodal scheme is listed in table 1.

Table 1: Explicit update rules by node for the five-node mesh.

Node i	Role	Update equation
0	Centre (symmetry)	eq. (20) (factor 6)
1, 2, 3	Interior	eq. (11) with $1/i = 1, \frac{1}{2}, \frac{1}{3}$
4	Surface (insulated)	eq. (23) (factor 2)

5 Source Terms and Diffusion-Coefficient Models

The base model of section 2 describes passive redistribution of an existing species. In-pile conditions add generation and loss processes. Each enters the governing equation as an additive term on the right-hand side,

$$\frac{\partial c}{\partial t} = D \left(\frac{\partial^2 c}{\partial r^2} + \frac{2}{r} \frac{\partial c}{\partial r} \right) + S(c, r, t), \quad (24)$$

and is discretised pointwise so that S is evaluated at node i and time level j before the explicit update.

5.1 Fission-product generation

Fresh fission products are created wherever fission occurs, at a volumetric rate

$$S_{\text{gen}} = \Sigma_f \phi Y, \quad (25)$$

where Σ_f is the macroscopic fission cross section of the kernel, ϕ is the local neutron flux, and Y is the fission-product yield of the species of interest. This term is positive and, for a flat flux across the small kernel, nearly uniform, so it raises the whole profile at a steady rate.

5.2 Radioactive decay

A radioactive species is lost in proportion to its local concentration,

$$S_{\text{decay}} = -\lambda c, \quad \lambda = \frac{\ln 2}{t_{1/2}}, \quad (26)$$

with λ the decay constant and $t_{1/2}$ the half-life. The term is linear in c and acts as a distributed sink that drives the concentration toward zero on the timescale $1/\lambda$.

5.3 Grain-boundary trapping

A fraction of the migrating fission products is immobilised at grain boundaries and defect sites and no longer participates in diffusion. This is represented by a linear trapping sink,

$$S_{\text{trap}} = -\kappa c, \quad (27)$$

where κ is the net rate at which species are trapped and rendered immobile. Like decay, it is linear in c , but it transfers material to an immobile population rather than removing it physically, which matters when a trapped inventory is tracked separately.

5.4 Diffusion-coefficient models

Two base cases are considered for the coefficient D .

Base case 1 (constant D). The diffusion coefficient is held fixed. This isolates the numerical behaviour of the scheme and admits comparison with closed-form solutions of the linear diffusion equation.

Base case 2 (temperature-dependent D). The coefficient follows an Arrhenius dependence on temperature T ,

$$D(T) = D_0 \exp\left(-\frac{Q}{R_g T}\right), \quad (28)$$

with pre-exponential factor D_0 , activation energy Q , and gas constant R_g . A radial or time-varying temperature field then makes D a function of position and time, and it must be evaluated inside the spatial loop alongside the curvature weight.

5.5 Further considerations

Two effects are noted as extensions beyond the present scope. First, D differs between the distinct coating layers of a TRISO particle for a given species, so a multilayer model assigns a separate coefficient to each material region and enforces flux continuity at the interfaces. Second, D evolves with accumulated irradiation damage, which makes the coefficient a function of burnup or fluence and couples the transport problem to the damage history of the kernel.

6 Verification Strategy and Planned Numerical Experiments

This section defines the computational experiments and the criteria by which the solver will be judged. The numerical implementation and its output are reported separately; the framework below fixes what will be computed and what a correct result looks like.

6.1 Base experiment

The primary experiment marches eq. (10) with the nodal rules of table 1 from a prescribed initial profile, with the source terms of section 5 switched off and the diffusion coefficient held constant (Base case 1). The concentration is recorded at every node $i = 0, \dots, 4$ at each time

level and the radial profiles are overlaid across successive time steps. The expected behaviour is monotone relaxation of the profile toward the state consistent with the imposed boundary condition, without spurious oscillation. Figure 1 is reserved for the overlaid concentration profiles and fig. 2 for the time history of the centre node. Captions state the mesh, the Fourier number, and the boundary type used in each run.

Figure 1: Radial concentration profiles C_i^j for $i = 0, \dots, 4$ overlaid at successive time levels for the constant- D , source-free base case. Placeholder: figure to be generated from the solver output.

Figure 2: Time history of the centre-node concentration C_0^j showing relaxation toward equilibrium. Placeholder: figure to be generated from the solver output.

6.2 Verification cases

The solver will be benchmarked against cases with known behaviour, following the strategy of focusing first on the source-free base case and then adding complexity.

1. **Conservation.** With insulated boundaries (eqs. (20) and (23)) and no source terms, the volume-weighted total inventory must remain constant to within rounding. Any drift signals an error in the curvature weighting or the boundary stencils.
2. **Approach to uniformity.** From any initial profile, an insulated kernel must relax to a spatially uniform concentration equal to the volume average of the initial state. The steady profile provides a direct check on the interior $1/i$ weights.
3. **Analytical limit.** For a constant coefficient and a Dirichlet surface value, the transient is compared against the series solution of the spherical diffusion equation, and the discrete error is monitored as Δr and Δt are refined.
4. **Stability boundary.** The Fourier number of eq. (12) is swept across $\text{Fo} = \frac{1}{2}$ to confirm that the explicit scheme remains bounded below the limit and develops growing oscillations above it.

6.3 Source-term and coefficient studies

Once the base case is verified, the generation, decay, and trapping terms of section 5 are activated individually and in combination, and Base case 2 with a temperature-dependent coefficient is run to assess the effect of an Arrhenius $D(T)$ on the release profile. Each addition is checked against the limiting behaviour it is expected to produce, for example a uniform upward drift under pure generation and an exponential decay of the total inventory under pure radioactive loss.

7 Discussion

7.1 Flux-conservative grouping

The interior stencil of eq. (9) can be regrouped so that the neighbour contributions carry their geometric weights explicitly. Combining the second and first difference terms over the common denominator $r \Delta r^2$ gives

$$\frac{\partial c}{\partial t} = D \left[\frac{C_{i-1}(r - \Delta r) + C_i(-2r) + C_{i+1}(r + \Delta r)}{r \Delta r^2} \right]. \quad (29)$$

In this grouping the inner neighbour is weighted by $(r - \Delta r)$ and the outer neighbour by $(r + \Delta r)$, which makes the curvature bias of spherical geometry transparent: the larger outer shell exchanges more material than the smaller inner shell. The form is algebraically identical to eq. (10) and is useful both as an implementation cross-check and as the starting point for an implicit scheme.

7.2 Toward an implicit scheme

The explicit update is inexpensive but conditionally stable, which forces small time steps when D is large or Δr is small. Evaluating the right-hand side at the new time level instead converts eq. (29) into a backward-Euler update of the general implicit form

$$x^{n+1} = x^n + h f(x^{n+1}, t^{n+1}), \quad (30)$$

in which the unknown C_i^{j+1} appears on both sides and is obtained by making C_{i+1}^{j+1} the subject and solving the resulting tridiagonal system across the mesh. Averaging the explicit and implicit right-hand sides produces the Crank-Nicolson scheme [?], which is second-order accurate in time and unconditionally stable for the linear diffusion operator [?]. The trade is a linear solve at each step in exchange for freedom from the Fourier-number restriction of eq. (12).

7.3 Limitations

Several limitations follow from the modelling choices. The coarse five-node mesh used for illustration resolves only the gross shape of the profile and would be refined for quantitative work. The explicit scheme is first-order in time and second-order in space, so accuracy is set by the smaller of the two truncation errors. The single-region treatment ignores the distinct coating layers of a real TRISO particle and the flux continuity that must hold at their interfaces. The constant-coefficient base case omits the temperature and irradiation dependence of D , both of which are significant over a realistic burnup history. Finally, the uniform-source and surface-coupled assumptions of section 2 suppress any angular structure in the kernel, which is acceptable for a symmetric particle but not for one with localised damage or asymmetric loading. These limitations are deliberate. They keep the baseline auditable and give a clear sequence of extensions, namely mesh refinement, an implicit time integrator, multilayer coupling, and a temperature- and damage-dependent coefficient, each of which can be verified against the base case before it is trusted.

8 Conclusion

A compact model of fission-product diffusion in a spherical TRISO kernel has been formulated and discretised. Starting from Fick’s second law, the assumption of shell uniformity reduces the governing equation to the one-dimensional radial form of eq. (5), in which a planar diffusion term is supplemented by a curvature term weighted by $1/r$. An FTCS finite-difference scheme converts this into the explicit nodal update of eq. (10), with the centre singularity removed through the L’Hopital limit that yields the $3D \partial^2 c / \partial r^2$ operator and a factor-of-six centre update, while the insulated surface, sitting at a finite radius where no singularity arises, carries an ordinary factor-of-two stencil; both end-node updates are collected in eqs. (20) and (23). The framework accommodates fission-product generation, radioactive decay, and grain-boundary trapping as additive terms, and supports both constant and Arrhenius-type diffusion coefficients. A verification programme has been defined around conservation, relaxation to uniformity, comparison with analytical limits, and the explicit stability boundary. The natural next steps are an implicit Crank-Nicolson integrator to lift the time-step restriction, refinement of the mesh, and extension to the multilayer, temperature- and damage-dependent coefficients characteristic of a full TRISO particle. The model in its present form provides the verified baseline on which those extensions can be built.

References